

— SAME PROMISE. DIFFERENT CONDITIONS.

+55.8%

FASTER

Controlled, self-contained task

Peng et al. • Microsoft Research / arXiv • 2023

-19%

SLOWER

Experienced developers, mature
repositories

METR • Mature repository study • 2025

Both results are real. What engineering conditions explain the difference?

— COGNITIONX EGYPT · 3 AUGUST 2026

AI-Native Development

From Writing Code to Owning the Full SDLC

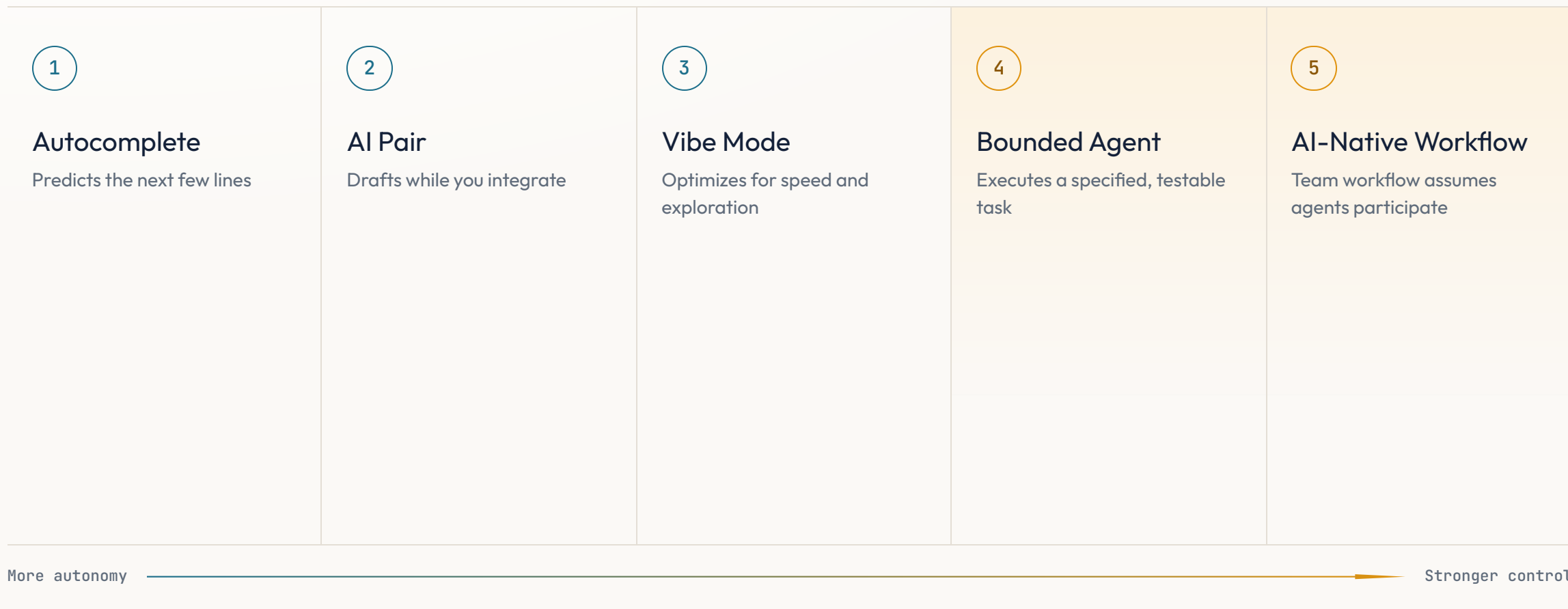
You are the orchestrator—not the typist.

Direction · Constraints · Evidence · Outcomes

Hany Saad Senior Engineering Manager, ITWorx

— OPERATING MODES, NOT A MATURITY LADDER

Five Modes of AI Development

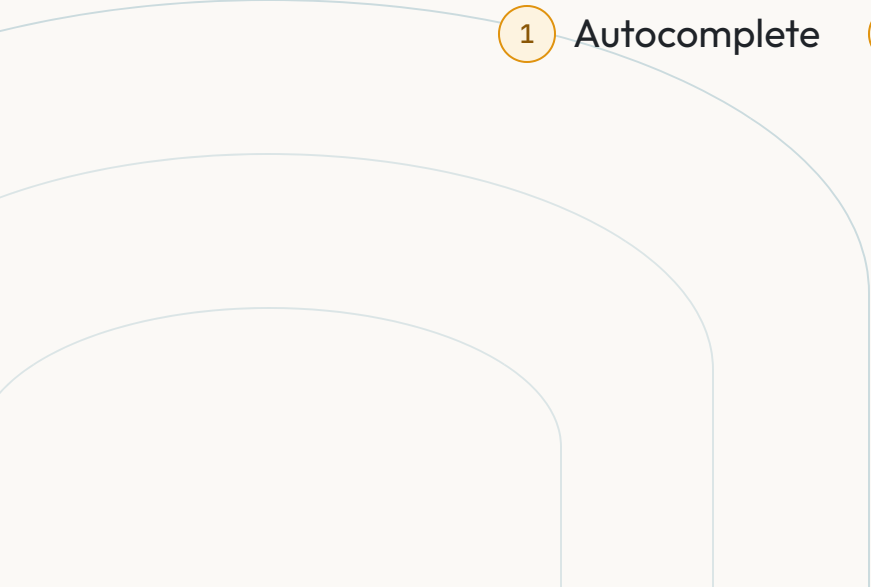


More autonomy requires stronger specification, boundaries, evidence, and accountability.

Every mode defines: delegated scope · human involvement · required controls · accountability

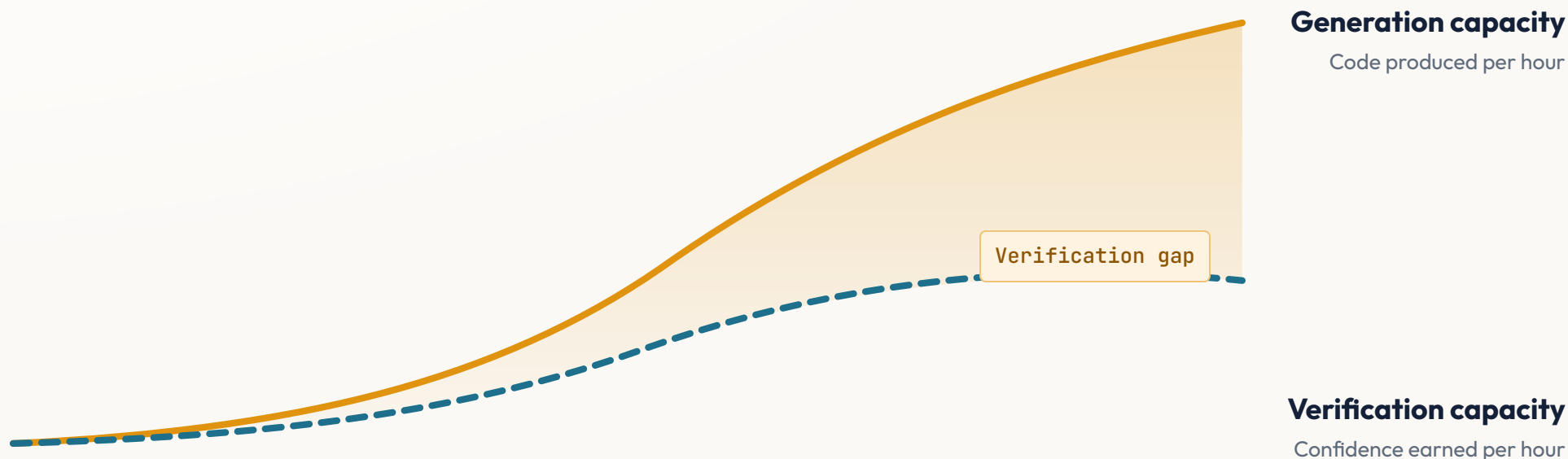
Where Are You Today?

Pick the mode you worked in most this past week.

- 
- 1 Autocomplete
 - 2 AI Pair
 - 3 Vibe
 - 4 Bounded Agent
 - 5 AI-Native Team

— THE NEW BOTTLENECK IS CONFIDENCE

The Verification Gap

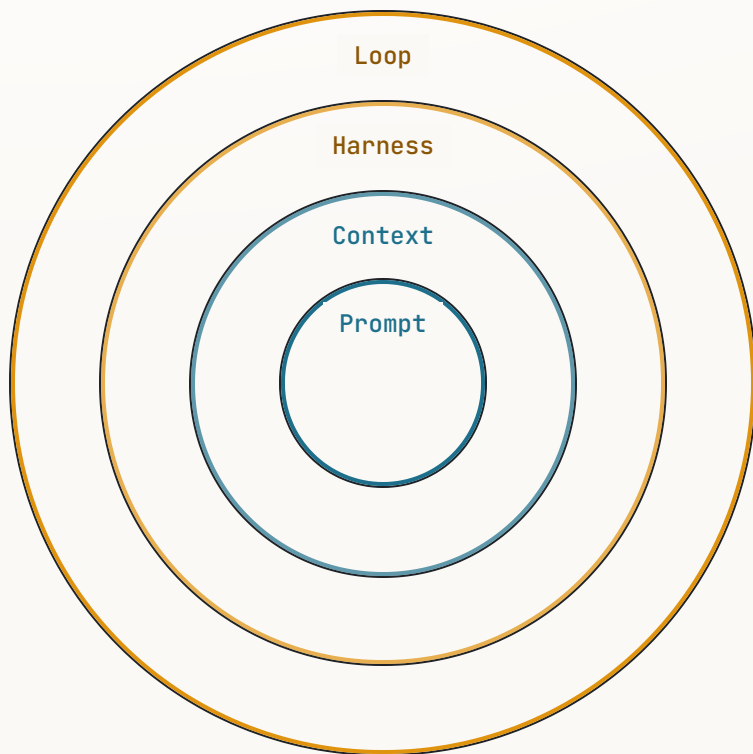


Faster output does not automatically mean faster delivery.

review effort • rework • escaped defects • delivery instability • trust

DORA • State of AI-assisted Software Development • 2025

Prompt → Context → Harness → Loop



- | | | |
|---|----------------|--|
| 1 | Prompt | What are we asking for? |
| 2 | Context | What does the agent need to know? |
| 3 | Harness | What can it access, execute, and change? |
| 4 | Loop | How does it verify, retry, stop, and escalate? |

Each layer contains—and depends on—the one before it.

— RESPONSIBILITY EXPANDS WITH LEVERAGE

Three Loops. Three Clocks.



The inner loop got fast. Your job moved outward.

— THE LIFECYCLE REMAINS

The SDLC Stayed. The Verbs Changed.

LIFECYCLE STAGE	TRADITIONAL EMPHASIS	AI-NATIVE EMPHASIS
Discovery	Gather	Frame
Requirements	Document	Specify
Design	Diagram	Constrain
Implementation	Type	Delegate
Testing	Validate afterwards	Define evidence first
Review	Read the diff	Assure the change
Release	Ship	Prove and recover

AI changes where engineers apply judgment—not whether judgment is needed.

Critical and high-risk code still receives code-level inspection.

What Owning the Full SDLC Actually Means

- | | | |
|----|----------------------------|---|
| 01 | Intent | Problem, outcome, and the decision not to build |
| 02 | Specification | Acceptance criteria and edge cases |
| 03 | Architecture | Boundaries, contracts, and data flows |
| 04 | Delegation boundary | Scope, access, tools, and stopping conditions |
| 05 | Evidence pack | Tests, security, performance, and observability |
| 06 | Release case | Rollout, provenance, and rollback |
| 07 | Production learning | Telemetry, incidents, feedback, and improvement |

Every artifact has an owner. AI is not that owner.

Start With Intent, Not Code

“Prevent session overbooking. If capacity is reached, offer a waitlist and notify organizers.”

Raw business request

What counts as capacity?

Who may override it?

What personal data is exposed?

What is the rollback behavior?

Is check-in concurrent?

How is the waitlist ordered?

What happens when notification fails?

ACCEPTANCE CRITERIA

Atomic capacity check

Authorized override

Stable waitlist order

Failure-tolerant notification

Auditable rollback

The engineer creates value before the first line is generated.

— CHALLENGE THE PLAN BEFORE ACCEPTING THE WORK

Plan. Constrain. Delegate.

AGENT'S PROPOSED PLAN

- 01 Add waitlist storage
- 02 Update check-in endpoint
- 03 Add organizer notification
- 04 Update attendee UI
- 05 Add tests

HUMAN REVIEW

- Concurrency is missing
- Authorization boundary is unclear
- Notification failure must not reverse check-in
- No new dependency without approval
- Audit event required

Delegation contract bounded tasks · affected interfaces · tests required · permissions allowed · stop / escalation conditions

— THE EVIDENCE IS THE DEMO

Verify the System, Not the Diff

01 Capacity acceptance tests	02 Concurrent check-in test	03 Authorization test	04 Migration and rollback check
05 Telemetry and alert	06 Dependency and secret scan	07 Manual review of critical logic	
Requirement satisfied	Risks addressed	Evidence attached	Ready for controlled release

Explainable. Testable. Observable. Reversible.

Code Review Is Necessary. It Is No Longer Sufficient.

01 Business intent

02 Acceptance behavior

03 Interfaces and data contracts

04 Architecture and dependencies

05 Security and supply chain

06 Operability and rollback

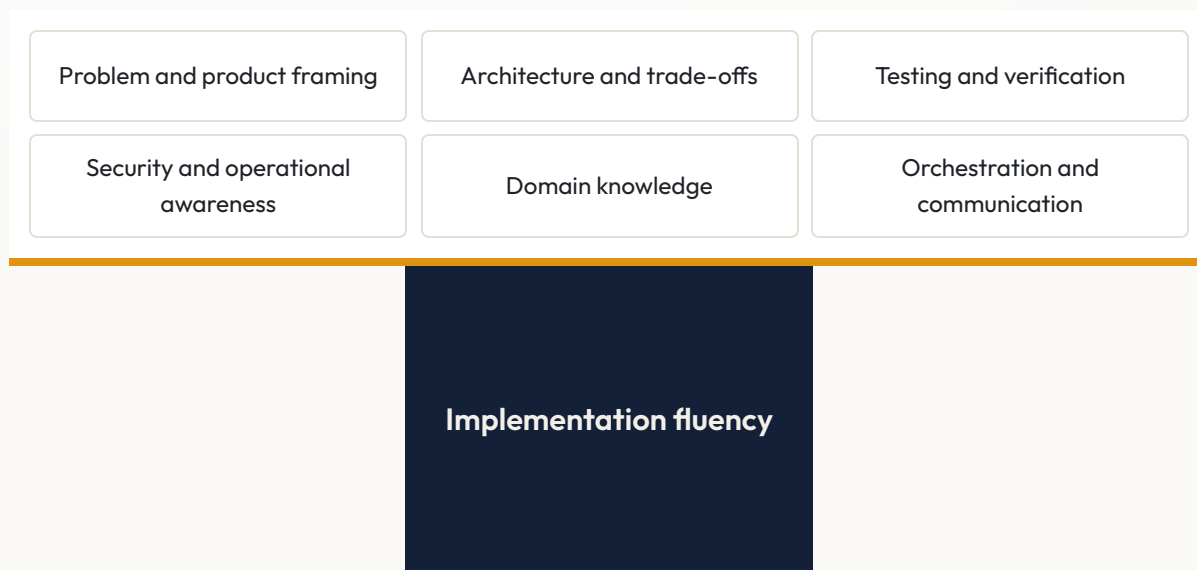
07 Critical code

Do not stop reviewing code.

Stop pretending code review alone proves the system is safe.

— IMPLEMENTATION FLUENCY REMAINS ESSENTIAL

The Engineer Moves Up the Stack



Junior Learn concepts and critique generated work earlier

Senior Scale judgment across more delivery

Architect Make boundaries machine-readable

Manager Redesign workflow, controls, and measurement

Skills once concentrated in senior roles are becoming earlier-career differentiators.

Monday Morning: Five Moves

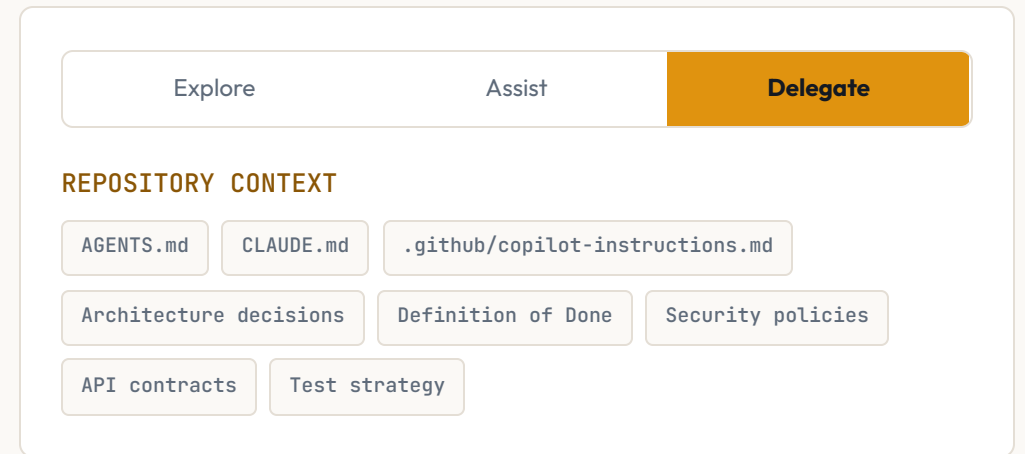
1 Choose one bounded, real task

2 Write acceptance criteria before invoking the agent

3 Make repository context explicit

4 Put automated evidence before human attention

5 Measure delivery—not generated lines



MEASURE DELIVERY

cycle time

review effort

rework

escaped defects

change failure rate

— OWN THE OUTER LOOPS

You are the orchestrator—not the typist.

The inner loop got fast. Own the outer loops.

AI-native development is not about producing more code.

It is about owning the system that turns intent into reliable outcomes.



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